STAT 6900: Stochastic Processes in Modern Machine Learning
Lecture 4: Stochastic Gradient Descent and Diffusion Approximation

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Notation and SGD Setup

We consider the finite-sum objective

$$\min_{x \in \mathbb{R}^d} f(x) := \frac{1}{n} \sum_{i=1}^{n} f_i(x).$$

SGD iteration:

$$x_{k+1} = x_k - \eta \nabla f_{\gamma_k}(x_k), \qquad k \geq 0, \tag{1}$$

where $\{\gamma_k\}$ are i.i.d. uniform on $\{1, 2, \ldots, n\}$ and η is the learning rate.

Indexing convention. Sequential index is k; dimensional coordinate uses brackets:

$$x_{k,(i)} \text{ is the } i\text{th coordinate of } x_k.$$

Rewriting SGD: Drift + Fluctuations

Rewrite the SGD iteration as

$$x_{k+1} - x_k = -\eta \nabla f(x_k) + \sqrt{\eta}\, V_k, \tag{2}$$

where

$$V_k := \sqrt{\eta}\big(\nabla f(x_k) - \nabla f_{\gamma_k}(x_k)\big).$$

Conditioned on x_k, we have

$$\mathbb{E}[V_k \mid x_k] = 0.$$

This decomposition separates:

- a *deterministic drift* term $-\eta \nabla f(x_k)$,
- a *random fluctuation* term $\sqrt{\eta}\, V_k$.

Noise Covariance $\Sigma(x)$

Conditioned on x_k, the covariance of V_k is

$$\mathrm{Cov}(V_k \mid x_k) = \eta\, \Sigma(x_k),$$

where the matrix-valued function $\Sigma : \mathbb{R}^d \to \mathbb{R}^{d\times d}$ is

$$\Sigma(x) = \frac{1}{n} \sum_{i=1}^{n} \big(\nabla f(x) - \nabla f_i(x)\big)\big(\nabla f(x) - \nabla f_i(x)\big)^\top. \tag{3}$$

Thus the stochasticity enters through the *state-dependent* matrix $\Sigma(x)$.

Candidate Continuous-Time Approximation

Consider the SDE

$$dX_t = b(X_t)\,dt + \sigma(X_t)\,dW_t, \qquad X_0 = x_0, \tag{4}$$

and compare with its Euler discretization (with step Δt):

$$X_{k+1} = X_k + \Delta t\, b(X_k) + \sqrt{\Delta t}\,\sigma(X_k) Z_k, \qquad Z_k \sim N(0, I).$$

If we identify $\Delta t = \eta$, then (2) suggests choosing

$$b \approx -\nabla f, \qquad \sigma \approx (\eta\Sigma)^{1/2},$$

and interpreting time by

$$t = k\eta.$$

Why "Weak" Approximation?

The driving noises of SGD and the SDE are *different* random mechanisms.
So we cannot compare sample paths directly (pathwise/strong sense).
Instead, we compare *distributions* of iterates at matched times.

This motivates the following notion of weak approximation used in numerical analysis of SDEs.

Definition 1: Order-α Weak Approximation

Let $0 < \eta < 1$, $T > 0$, and set

$$N = \lfloor T/\eta \rfloor.$$

Let $\mathcal{G}$ denote the set of functions of polynomial growth:

$$g \in \mathcal{G} \quad \Longleftrightarrow \quad \exists\, K, \kappa > 0 \text{ s.t. } |g(x)| < K(1 + |x|^{\kappa}).$$

We say that the SDE (4) is an *order α weak approximation* to SGD (1) if for every $g \in \mathcal{G}$, there exists $C > 0$ independent of η such that for all $k = 0, 1, \ldots, N$,

$$\left|\mathbb{E}g(X_{k\eta}) - \mathbb{E}g(x_k)\right| \; < \; C\,\eta^{\alpha}.$$

Informal Statement of Theorem 1

Assume f, f_i are Lipschitz, have at most linear growth, and have sufficiently high derivatives belonging to $\mathcal{G}$. Let Σ be defined by (3).

Then, for fixed T:

- The SDE
$$dX_t = -\nabla f(X_t)\,dt + (\eta\Sigma(X_t))^{1/2}dW_t \tag{5}$$
is an *order 1* weak approximation of SGD.
- The SDE
$$dX_t = -\nabla\Big(f(X_t) + \frac{\eta}{4}\,|\nabla f(X_t)|^2\Big)dt + (\eta\Sigma(X_t))^{1/2}dW_t \tag{6}$$
is an *order 2* weak approximation of SGD.

First-Order SME (Order 1 Weak)

First-order stochastic modified equation (SME):

$$dX_t = -\nabla f(X_t)\,dt + (\eta\Sigma(X_t))^{1/2}\,dW_t, \qquad X_0 = x_0.$$

Key points (as a lecture takeaway):

- Drift matches the full-gradient flow $-\nabla f$.
- Diffusion magnitude is $O(\eta^{1/2})$ through $(\eta\Sigma)^{1/2}$.
- Approximation is *distributional* with weak error $O(\eta)$.

Second-Order SME (Order 2 Weak)

Second-order SME:

$$dX_t = -\nabla\Big(f(X_t) + \frac{\eta}{4}|\nabla f(X_t)|^2\Big)\,dt + (\eta\Sigma(X_t))^{1/2}\,dW_t, \qquad X_0 = x_0.$$

Interpretation of the correction term:

$$-\nabla\Big(\frac{\eta}{4}|\nabla f(X_t)|^2\Big)$$

is an additional drift of size $O(\eta)$ that improves weak accuracy from order 1 to order 2.

This is the SME used for *exact calculations* later in the dynamics section.

Formal Theorem 1 (Stochastic Modified Equations)

Let $\alpha \in \{1, 2\}$, $0 < \eta < 1$, $T > 0$, and set $N = \lfloor T/\eta \rfloor$ and $\{x_k\}_{k=0}^{N}$ be SGD iterates defined by (1). Define X_t by the SDE

$$dX_t = -\nabla\Big(f(X_t) + \frac{1}{4}(\alpha - 1)\eta|\nabla f(X_t)|^2\Big)\,dt + (\eta\Sigma(X_t))^{1/2}dW_t, \quad (7)$$

with $X_0 = x_0$ and $\Sigma(x)$ given by (4).
Fix $g \in \mathcal{G}$ and assume:

(i) (**Lipschitz**) $\exists L > 0$ s.t.

$$|\nabla f(x) - \nabla f(y)| + \sum_{i=1}^{n} |\nabla f_i(x) - \nabla f_i(y)| \leq L|x - y|.$$

(ii) f, f_i and partial derivatives up to order 7 belong to $\mathcal{G}$.

(iii) (**Growth**) $\exists M > 0$ s.t.

$$|\nabla f(x)| + \sum_{i=1}^{n} |\nabla f_i(x)| \leq M(1 + |x|).$$

(iv) g and partial derivatives up to order 6 belong to $\mathcal{G}$.

Formal Theorem 1 (Stochastic Modified Equations)

Then $\exists C > 0$ independent of η s.t. for all $k = 0, 1, \ldots, N$,

$$\left|\mathbb{E}g(X_{k\eta}) - \mathbb{E}g(x_k)\right| \leq C\,\eta^{\alpha}.$$

That is, the equation (7) is an order α weak approximation of the SGD iteration.

A Solvable Example

Example 4.1

We consider the special case:

$$n = 2, \qquad d = 1.$$

Define

$$f_1(x) = x^2, \qquad f_2(x) = -x^2.$$

Then the objective function is

$$f(x) = \frac{1}{2}(f_1(x) + f_2(x)) = 0.$$

So the deterministic gradient vanishes:

$$\nabla f(x) = 0.$$

All dynamics arise from stochastic fluctuations.

Last Example (cont.)

Example 4.1

Since γ_k is uniform on $\{1, 2\}$,

$$\nabla f_{\gamma_k}(x) = \begin{cases} 2x, & \text{with prob. } 1/2, \\ -2x, & \text{with prob. } 1/2. \end{cases}$$

The SGD update is

$$x_{k+1} = x_k - \eta \nabla f_{\gamma_k}(x_k).$$

Hence,

$$x_{k+1} = \begin{cases} (1 - 2\eta)x_k, & \text{with prob. } 1/2, \\ (1 + 2\eta)x_k, & \text{with prob. } 1/2. \end{cases}$$

Last Example (cont.)

Example 4.1

Since $\nabla f(x) = 0$, equation (3) reduces to

$$x_{k+1} - x_k = \sqrt{\eta} V_k.$$

From the definition

$$V_k = \sqrt{\eta}\big(\nabla f(x_k) - \nabla f_{\gamma_k}(x_k)\big),$$

we obtain

$$V_k = \begin{cases} -2\sqrt{\eta}\, x_k, & \text{with prob. } 1/2, \\ 2\sqrt{\eta}\, x_k, & \text{with prob. } 1/2. \end{cases}$$

Thus the noise is multiplicative.

Last Example (cont.)

Example 4.1

From equation (3),

$$\Sigma(x) = \frac{1}{2}\left[(0-2x)^2 + (0+2x)^2\right].$$

Therefore,

$$\Sigma(x) = 4x^2.$$

The first-order SME (5) becomes

$$dX_t = (\eta\Sigma(X_t))^{1/2} dW_t.$$

Substitute $\Sigma(x) = 4x^2$:

$$dX_t = 2\sqrt{\eta}\, X_t\, dW_t.$$

Last Example (cont.)

Example 4.1

We obtain the SDE

$$dX_t = 2\sqrt{\eta}\, X_t\, dW_t, \qquad X_0 = x_0. \tag{8}$$

This is a geometric Brownian motion with zero drift.

It admits the explicit solution

$$X_t = x_0 \exp\Big(2\sqrt{\eta}\, W_t - 2\eta t\Big).$$

Last Example (cont.)

Example 4.1

Using properties of geometric Brownian motion,

$$\mathbb{E}[X_t] = x_0.$$

Thus the mean remains constant in time.

Even though individual trajectories fluctuate, there is no deterministic descent.

Last Example (cont.)

Example 4.1

Compute:

$$\mathbb{E}[X_t^2] = x_0^2 \exp(4\eta t).$$

Hence

$$\mathrm{Var}(X_t) = x_0^2 \left(e^{4\eta t} - 1\right).$$

Variance grows exponentially.

Last Example (cont.)

Example 4.1

In this Example 4.1:

- Deterministic gradient is zero.
- All dynamics are noise-driven.
- Noise is multiplicative: magnitude proportional to X_t.
- Fluctuations amplify over time.

This shows that SGD can exhibit instability even when $\nabla f = 0$.

A Solvable Example with True Descent

Example 4.2

We keep $n = 2$, $d = 1$, and define

$$f_1(x) = x^2, \qquad f_2(x) = x^2.$$

Then

$$f(x) = \frac{1}{2}(f_1(x) + f_2(x)) = x^2, \qquad \nabla f(x) = 2x.$$

SGD with $\gamma_k \sim \mathrm{Unif}\{1, 2\}$ becomes deterministic here, so this example alone has no noise.

To obtain a nontrivial solvable model with both drift and noise, we use the paper's construction where $\Sigma(x)$ is nonzero while $f(x) = x^2$ produces descent.

Last Example (cont.)

Example 4.2

Recall that the first-order SME (5) is

$$dX_t = -\nabla f(X_t)\,dt + (\eta\Sigma(X_t))^{1/2}\,dW_t.$$

In the one-dimensional quadratic case

$$f(x) = x^2 \quad \Rightarrow \quad \nabla f(x) = 2x,$$

and the noise structure is chosen so that

$$\Sigma(x) = 4x^2,$$

which yields a closed-form SDE.

Thus the SME becomes

$$dX_t = -2X_t\,dt + 2\sqrt{\eta}\,X_t\,dW_t, \qquad X_0 = x_0. \tag{9}$$

Last Example (cont.)

Example 4.2

The SDE (9) is a geometric Brownian motion with drift.

Applying Itô to $\log X_t$ (for $X_t > 0$) gives

$$d(\log X_t) = \left(-2 - \frac{1}{2}(2\sqrt{\eta})^2\right) dt + 2\sqrt{\eta}\, dW_t = -2(1+\eta)\, dt + 2\sqrt{\eta}\, dW_t.$$

Therefore,

$$X_t = x_0 \exp\Big(-2(1+\eta)t + 2\sqrt{\eta}\, W_t \Big). \tag{10}$$

Last Example (cont.)

Example 4.2

Using $\mathbb{E}[e^{aW_t}] = e^{a^2 t/2}$,

$$\mathbb{E}[X_t] = x_0 \exp\Big(-2(1+\eta)t\Big)\,\mathbb{E}\big[e^{2\sqrt{\eta}W_t}\big] = x_0 \exp\Big(-2(1+\eta)t\Big)\exp(2\eta t).$$

Thus,

$$\mathbb{E}[X_t] = x_0 e^{-2t}.$$

So the mean follows the deterministic gradient flow for $f(x) = x^2$.

Last Example (cont.)

Example 4.2

Compute

$$X_t^2 = x_0^2 \exp\Big(-4(1+\eta)t + 4\sqrt{\eta}\, W_t \Big).$$

Then

$$\mathbb{E}[X_t^2] = x_0^2 \exp\Big(-4(1+\eta)t \Big)\, \mathbb{E}\big[e^{4\sqrt{\eta}W_t}\big] = x_0^2 e^{-4(1+\eta)t} \exp\Big(\frac{(4\sqrt{\eta})^2}{2} t \Big).$$

Hence,

$$\mathbb{E}[X_t^2] = x_0^2 \exp\big((-4 - 4\eta + 8\eta)t \big) = x_0^2 e^{(-4+4\eta)t}.$$

Last Example (cont.)

Example 4.2

We have

$$\mathrm{Var}(X_t) = \mathbb{E}[X_t^2] - (\mathbb{E}[X_t])^2 = x_0^2 e^{(-4+4\eta)t} - x_0^2 e^{-4t}.$$

Factor out $x_0^2 e^{-4t}$:

$$\mathrm{Var}(X_t) = x_0^2 e^{-4t}\Big(e^{4\eta t} - 1\Big).$$

This expression cleanly separates:

- deterministic descent: e^{-4t},
- stochastic amplification: $e^{4\eta t} - 1$.

Last Example (cont.)

Example 4.2

Consider the relative fluctuation size:

$$\frac{\sqrt{\mathrm{Var}(X_t)}}{|\mathbb{E}[X_t]|} = \frac{\sqrt{x_0^2 e^{-4t}(e^{4\eta t} - 1)}}{|x_0| e^{-2t}} = \sqrt{e^{4\eta t} - 1}.$$

Thus the ratio depends only on ηt.

For small ηt,

$$e^{4\eta t} - 1 \approx 4\eta t, \qquad \frac{\sqrt{\mathrm{Var}(X_t)}}{|\mathbb{E}[X_t]|} \approx \sqrt{4\eta t}.$$

For large ηt, this ratio grows like $e^{2\eta t}$.

Last Example (cont.)

Example 4.2

A natural transition is when fluctuations become comparable to the mean:

$$\sqrt{\mathrm{Var}(X_t)} \sim \big|\mathbb{E}[X_t]\big|.$$

Using

$$\frac{\sqrt{\mathrm{Var}(X_t)}}{|\mathbb{E}[X_t]|} = \sqrt{e^{4\eta t} - 1},$$

the transition condition becomes

$$e^{4\eta t} \sim 2.$$

Therefore,

$$t^* = \frac{\log 2}{4\eta}.$$

For $t \ll t^*$: descent dominates. For $t \gg t^*$: fluctuations dominate.

Last Example (cont.)

Example 4.2

The SME

$$dX_t = -2X_t\,dt + 2\sqrt{\eta}\,X_t\,dW_t$$

has:

- mean $x_0 e^{-2t}$ (pure descent),
- variance $x_0^2 e^{-4t}(e^{4\eta t} - 1)$ (descent $\times$ amplification),
- transition time $t^* = \frac{\log 2}{4\eta}$.

This makes the role of the learning rate η explicit: it controls how quickly stochastic fluctuations overwhelm descent.

Solvable SME: Exact Quadratic Construction

Example 4.3

Let $n = 2$, $d = 1$, and set

$$f(x) = x^2, \qquad f_1(x) = (x-1)^2 - 1, \qquad f_2(x) = (x+1)^2 - 1.$$

Then indeed

$$\frac{1}{2}\big(f_1(x) + f_2(x)\big) = x^2 = f(x).$$

We will compute ∇f and $\Sigma(x)$ ((3)) explicitly, then plug into the second-order SME ((6)).

Last Example (cont.)

Example 4.3

Compute gradients (here, derivatives):

$$\nabla f(x) = 2x, \qquad \nabla f_1(x) = 2(x-1), \qquad \nabla f_2(x) = 2(x+1).$$

Hence the gradient discrepancies are

$$\nabla f(x) - \nabla f_1(x) = 2x - 2(x-1) = 2,$$

$$\nabla f(x) - \nabla f_2(x) = 2x - 2(x+1) = -2.$$

So the noise magnitude is constant (independent of x).

Last Example (cont.)

Example 4.3

By (3),

$$\Sigma(x) = \frac{1}{2}\sum_{i=1}^{2}\big(\nabla f(x) - \nabla f_i(x)\big)^2.$$

Using the previous slide:

$$\Sigma(x) = \frac{1}{2}\big(2^2 + (-2)^2\big) = 4.$$

Thus

$$\Sigma(x) \equiv 4 \quad \text{(constant)}.$$

Last Example (cont.)

Example 4.3

Recall (6):

$$dX_t = -\nabla\Big(f(X_t) + \frac{\eta}{4}\,|\nabla f(X_t)|^2\Big)\,dt + (\eta\Sigma(X_t))^{1/2}dW_t.$$

Here, $f(x) = x^2$ and $\nabla f(x) = 2x$, so

$$|\nabla f(x)|^2 = 4x^2, \qquad f(x) + \frac{\eta}{4}|\nabla f(x)|^2 = x^2 + \eta x^2 = (1+\eta)x^2.$$

Therefore

$$\nabla\Big(f(x) + \frac{\eta}{4}|\nabla f(x)|^2\Big) = 2(1+\eta)x.$$

Last Example (cont.)

Example 4.3

Using $\Sigma(x) \equiv 4$, we have

$$(\eta\Sigma(X_t))^{1/2} = \sqrt{4\eta} = 2\sqrt{\eta}.$$

Hence the SME (Eq. (7)) becomes

$$dX_t = -2(1+\eta)X_t\,dt + 2\sqrt{\eta}\,dW_t, \qquad X_0 = x_0.$$

This is the (one-dimensional) Ornstein–Uhlenbeck process.

Last Example (cont.)

Example 4.3

For the OU process

$$dX_t = -2(1+\eta)X_t\,dt + 2\sqrt{\eta}\,dW_t, \qquad X_0 = x_0,$$

the distribution is Gaussian:

$$X_t \sim \mathcal{N}\Big(x_0 e^{-2(1+\eta)t},\ \frac{\eta}{1+\eta}\Big(1 - e^{-4(1+\eta)t}\Big)\Big).$$

So:

$$\mathbb{E}X_t = x_0 e^{-2(1+\eta)t}, \qquad \mathrm{Var}(X_t) = \frac{\eta}{1+\eta}\Big(1 - e^{-4(1+\eta)t}\Big).$$

Last Example (cont.)

Example 4.3

From the exact formulas:

$$\mathbb{E}X_t \to 0 \quad \text{exponentially with rate } 2(1+\eta),$$

while

$$\mathrm{Var}(X_t) \uparrow \frac{\eta}{1+\eta} \quad \text{as } t \to \infty.$$

Interpretation:

- **Descent phase:** mean dominates (trajectory contracts toward 0).
- **Fluctuation phase:** variance reaches an $O(\eta)$ plateau.

We define the separation time t^* by balancing these two effects.

Last Example (cont.)

Example 4.3

Define t^* by

$$\mathbb{E}X_{t^*} = \sqrt{\mathrm{Var}(X_{t^*})}.$$

Using the exact formulas:

$$x_0^2 e^{-4(1+\eta)t^*} = \frac{\eta}{1+\eta}\Big(1 - e^{-4(1+\eta)t^*}\Big).$$

Solving yields:

$$t^* = \frac{1}{4(1+\eta)} \log\Big(1 + \eta + \frac{1}{\eta x_0^2}\Big).$$

For $t < t^*$: descent dominates. For $t > t^*$: fluctuation dominates.

Last Example (cont.)

Example 4.3

In the identification between discrete SGD and continuous time,

$$t = k\eta.$$

Therefore the corresponding iteration index is

$$k^* = \frac{t^*}{\eta} = \frac{1}{4\eta(1+\eta)} \log\Big(1 + \eta + \frac{1}{\eta x_0^2}\Big).$$

This predicts where typical SGD sample paths stop looking like a clean contraction and start visibly fluctuating.

Stochastic Asymptotic Expansion: Motivation

We work with the first-order SME (for simplicity):

$$dX_t = -\nabla f(X_t)\,dt + (\eta\Sigma(X_t))^{1/2}\,dW_t, \qquad X_0 = x_0. \tag{6}$$

Key observation:

$$(\eta\Sigma(X_t))^{1/2} = O(\eta^{1/2}),$$

so the noise term is *small* when η is small.

Therefore, we can expand X_t in powers of $\eta^{1/2}$.

Introduce the Small Parameter

Set

$$\varepsilon := \sqrt{\eta}.$$

Then the SME (5) can be written as

$$dX_t^\varepsilon = b(X_t^\varepsilon)\,dt + \varepsilon\,\sigma(X_t^\varepsilon)\,dW_t, \tag{11}$$

where

$$b(x) := -\nabla f(x), \qquad \sigma(x) := \Sigma(x)^{1/2}.$$

We will derive the first two terms of the expansion in ε.

Asymptotic Series Ansatz

Assume an asymptotic series of stochastic processes:

$$X_t^\varepsilon = X_{0,t} + \varepsilon X_{1,t} + \varepsilon^2 X_{2,t} + \cdots . \tag{12}$$

Initial conditions are

$$X_{0,0} = x_0, \qquad X_{j,0} = 0 \;\; (j \geq 1).$$

We substitute (12) into (11) and match coefficients of powers of ε.

Taylor Expansions of Drift and Diffusion

Assuming smoothness, expand around $X_{0,t}$:

$$b(X_t^\varepsilon) = b(X_{0,t}) + \varepsilon\, \nabla b(X_{0,t})\, X_{1,t} + O(\varepsilon^2),$$

$$\sigma(X_t^\varepsilon) = \sigma(X_{0,t}) + \varepsilon\, \nabla \sigma(X_{0,t})\, X_{1,t} + O(\varepsilon^2).$$

Insert these into (11) and equate like powers of ε.

Leading-Order Equation: Deterministic Gradient Flow

At order ε^0 we obtain the ODE

$$dX_{0,t} = b(X_{0,t})\,dt.$$

With $b(x) = -\nabla f(x)$, this becomes

$$dX_{0,t} = -\nabla f(X_{0,t})\,dt, \qquad X_{0,0} = x_0.$$

Equivalently,

$$\dot{X}_{0,t} = -\nabla f(X_{0,t}).$$

First-Order Fluctuation Equation

At order ε^1, we obtain a linear SDE for $X_{1,t}$:

$$dX_{1,t} = \nabla b(X_{0,t})\, X_{1,t}\, dt + \sigma(X_{0,t})\, dW_t, \qquad X_{1,0} = 0. \tag{13}$$

Since $b(x) = -\nabla f(x)$,

$$\nabla b(X_{0,t}) = -H_f(X_{0,t}),$$

where H_f is the Hessian of f.
Thus,

$$dX_{1,t} = -H_f(X_{0,t})\, X_{1,t}\, dt + \Sigma(X_{0,t})^{1/2}\, dW_t. \tag{14}$$

Truncated Approximation

Define the truncated asymptotic approximation

$$\widehat{X}_t = X_{0,t} + \sqrt{\eta}\, X_{1,t}. \tag{15}$$

Interpretation:

- $X_{0,t}$ is the deterministic gradient-flow trajectory.
- $\sqrt{\eta}\, X_{1,t}$ captures leading stochastic fluctuations.

We now analyze the distribution of $\widehat{X}_t$.

Time-Dependent Linearization Along the Flow

Define the non-random, time-dependent matrices along $X_{0,t}$:

$$H_t := H_f(X_{0,t}), \qquad \Sigma_t := \Sigma(X_{0,t}). \tag{16}$$

Then (13) becomes the time-inhomogeneous linear SDE

$$dX_{1,t} = -H_t X_{1,t}\, dt + \Sigma_t^{1/2}\, dW_t, \qquad X_{1,0} = 0. \tag{17}$$

Since the drift is linear and the diffusion is independent of $X_{1,t}$, $X_{1,t}$ is Gaussian.

Mean of the Fluctuation Process

From (17), by Itô calculus,

$$\mathbb{E}[X_{1,t}] = 0.$$

Hence,

$$\widehat{\mathbb{E}}[\widehat{X}_t] = X_{0,t}.$$

So the mean of the approximation follows the gradient flow.

Covariance Dynamics: Lyapunov-Type ODE

Let

$$S_t := \mathrm{Cov}(X_{1,t}).$$

Then S_t satisfies the matrix ODE

$$\frac{d}{dt} S_t = -S_t H_t - H_t S_t + \Sigma_t, \qquad S_0 = 0.$$

This is a linearized version of a Riccati equation.

Special cases (closed-form possible):

- $d = 1$,
- H_t constant,
- commuting matrix families.

Gaussian Approximation of the SME Solution

Since $X_{1,t}$ is Gaussian with mean 0 and covariance S_t, the truncated approximation

$$\widehat{X}_t = X_{0,t} + \sqrt{\eta}\, X_{1,t}$$

is Gaussian with

$$\widehat{X}_t \sim \mathcal{N}(X_{0,t}, \eta S_t). \tag{18}$$

This provides an explicit approximation to the distribution of X_t (and hence to the SGD iterates, via the SME).

Two Regimes via S_t

From (18), the covariance scale is

$$\mathrm{Cov}(\widehat{X}_t) = \eta S_t.$$

Heuristic (using asymptotic stability of the linearized flow): if $H_t \to H_\infty$ and $\Sigma_t \to \Sigma_\infty$ as $t \to \infty$, then S_t approaches a steady state S_∞ with magnitude

$$|S_\infty| \sim \frac{|\Sigma_\infty|}{|H_\infty|}.$$

Hence, in the long-time fluctuating regime,

$$\mathrm{Cov}(\widehat{X}_t) \text{ is of order } O\left(\eta \frac{|\Sigma_\infty|}{|H_\infty|}\right).$$

Before that, dynamics is dominated by the gradient flow $X_{0,t}$.

Stochastic Asymptotic Expansion: Main Gaussian Approximation

From (18)

$$X_t \approx \mathcal{N}(X_{0,t}, \eta S_t).$$

Here:

$$\dot{X}_{0,t} = -\nabla f(X_{0,t}), \qquad X_{0,0} = x_0,$$

and S_t solves

$$\dot{S}_t = -S_t H_t - H_t S_t + \Sigma_t, \qquad S_0 = 0,$$

with

$$H_t = H_f(X_{0,t}), \qquad \Sigma_t = \Sigma(X_{0,t}).$$

This toll cab be used for predicting SGD dynamics when the SME is not explicitly solvable.

Truncation Example

Example 4.4

Recall that from Example (4.3)

$$f(x) = x^2 \;\Rightarrow\; H_f(x) = 2, \qquad \Sigma(x) \equiv 4.$$

Along the flow, $H_t \equiv 2$ and $\Sigma_t \equiv 4$ are constants.
Then the S_t-ODE becomes (scalar case):

$$\dot{S}_t = -S_t(2) - (2)S_t + 4 = -4S_t + 4, \qquad S_0 = 0,$$

so

$$S_t = 1 - e^{-4t}.$$

Thus (18) predicts $X_t \approx \mathcal{N}\big(X_{0;t}, \eta(1 - e^{-4t})\big)$, which matches the exact OU variance structure up to the $(1 + \eta)$ refinement that comes from using the second-order SME.

Adaptive Hyper-parameter Adjustment: Learning Rate Control

We now allow *learning-rate adjustment* through a control factor.

Discrete-time controlled iteration (1D):

$$x_{k+1} = x_k - \eta\, u_k\, f'(x_k), \qquad u_k \in [0, 1], \tag{19}$$

where η is the *maximum* allowed learning rate and u_k is an adjustment factor.

Continuous-time identification:

$$t = k\eta, \qquad u_t \text{ is the continuous analogue of } u_k.$$

Controlled SME: Drift and Noise Scale Together

The corresponding stochastic modified equation is

$$dX_t = -u_t f'(X_t)\,dt + u_t\sqrt{\eta\,\Sigma(X_t)}\,dW_t, \qquad u_t \in [0,1]. \tag{10}$$

Key structural point:

- Larger u_t increases the *drift magnitude* $\Rightarrow$ faster initial descent.
- The *same* u_t multiplies the noise term $\Rightarrow$ larger fluctuations.

Optimal Control Problem

Goal: choose an admissible control $u(\cdot)$ to minimize the final expected objective.

Control problem:

$$\min_u \ \mathbb{E}\, f(X_T) \quad \text{subject to} \quad \text{the controlled SME (10)},$$

where u ranges over Borel-measurable functions $u : [0, T] \to [0, 1]$.

Quadratic Objective with Constant Noise Example

Example 4.5

Assume

$$f(x) = \frac{1}{2}a(x-b)^2, \qquad a, b \in \mathbb{R},$$

and

$$\Sigma(x) = \Sigma > 0 \quad \text{(constant)}.$$

Then the controlled SME (10) becomes

$$dX_t = -au_t(X_t - b)\,dt + u_t\sqrt{\eta\Sigma}\,dW_t. \tag{20}$$

Reduce the Control Problem to a Scalar ODE

Define the controlled performance variable

$$m_t := \mathbb{E}\, f(X_t).$$

Apply Itô calculus to $f(X_t) = \frac{1}{2}a(X_t - b)^2$ under (20) and take expectation.

This yields the ODE:

$$\dot{m}_t = -2au_t m_t + \frac{1}{2}a\eta\Sigma\, u_t^2. \tag{21}$$

Thus the optimal control problem becomes:

$$\min_{u:[0,T]\to[0,1]} m_T \quad \text{subject to (21).}$$

Solving the ODE-Control Problem (HJB/Dynamic Programming)

We optimize the terminal cost m_T subject to the controlled dynamics (21).

The resulting optimal feedback control is:

$$u_t^* = \begin{cases} 1, & a \leq 0, \\ \min\Big(1,\ \dfrac{2m_t}{\eta\Sigma}\Big), & a > 0. \end{cases} \tag{22}$$

This is a *feedback law*: the action at time t depends on the current state variable m_t.

Interpretation: Concave Case $a < 0$

If $a < 0$ (concave objective), the policy is

$$u_t^* = 1.$$

Reason (mechanism in (21)):

- The drift term $-2au_t m_t$ is *helped* by taking u_t large.
- The noise term $\frac{1}{2}a\eta\Sigma u_t^2$ has $a < 0$, so noise-induced spread also decreases $\mathbb{E}f$.

Hence maximum learning rate is always optimal.

Interpretation: Convex Case $a > 0$

If $a > 0$ (convex objective), the policy is

$$u_t^* = \min\Big(1,\ \frac{2m_t}{\eta\Sigma}\Big).$$

Two regimes:

- If m_t is large relative to $\eta\Sigma$, then $u_t^* = 1$ (fast descent).
- When m_t becomes small, fluctuations dominate, and the policy decreases u_t proportionally to m_t.

So the feedback policy automatically transitions from *maximal step size* to *annealing*.

Explicit Annealing Schedule for $a > 0$

With u_t^* from (22), we can solve (21) and express u_t^* explicitly:

$$u_t^* = \begin{cases} 1, & a \leq 0 \text{ or } t \leq t^*, \\ \dfrac{1}{1 + a(t - t^*)}, & a > 0 \text{ and } t > t^*, \end{cases}$$

where

$$t^* = \frac{1}{2a} \log\Big(\frac{4m_0}{\eta\Sigma} - 1\Big).$$

After the onset of fluctuations, the optimal schedule decays like $\sim 1/t$.

Connection to the Descent–Fluctuation Transition

For the quadratic/constant-noise model, the time t^* marks the boundary between:

- **Descent phase:** $u_t^* = 1$ is optimal (maximize drift).
- **Fluctuation phase:** u_t^* decreases to control noise-induced variance.

In particular parameter choices recover the same qualitative transition as in the earlier OU solvable dynamics.

Extension to High Dimensions: Local Diagonal-Quadratic Model

For $d > 1$ and general f:
Assume locally (around current x),

$$f(x) \approx \frac{1}{2} \sum_{i=1}^{d} a^{(i)} \big(x^{(i)} - b^{(i)}\big)^2,$$

and

$$\Sigma(x) \approx \operatorname{diag}\{\Sigma^{(1)}, \ldots, \Sigma^{(d)}\} \quad \text{(locally constant on the diagonal).}$$

Then the control problem decouples across coordinates, so we apply the 1D feedback law (22) *element-wise*.

On-the-Fly Estimation via Local Linear Gradient Model

Using the local relation (coordinate-wise):

$$\partial^{(i)} f(x) \approx a^{(i)} \big(x^{(i)} - b^{(i)}\big),$$

we estimate $a^{(i)}$, $b^{(i)}$ by linear regression from online statistics.

Maintain exponential moving averages (EMA) per coordinate i:

$$\{\bar{g}_{k,(i)},\ \bar{g}^2_{k,(i)},\ \bar{x}_{k,(i)},\ \bar{x}^2_{k,(i)},\ \bar{xg}_{k,(i)}\}, \qquad \bar{g}_{k,(i)} = \partial^{(i)} f_{\gamma_k}(x_k).$$

These EMAs provide stabilized estimates of slope/intercept/variance.

Closed-Form Online Estimates for a_k, b_k, Σ_k

Define (coordinate-wise) the OLS-type estimates:

$$
\begin{aligned}
a_{k,(i)} &= \frac{\bar{gx}_{k,(i)} - \bar{g}_{k,(i)}\bar{x}_{k,(i)}}{\bar{x}^2_{k,(i)} - \bar{x}^2_{k,(i)}}, \\
b_{k,(i)} &= \bar{x}_{k,(i)} - \frac{\bar{g}_{k,(i)}}{a_{k,(i)}}, \\
\Sigma_{k,(i)} &= \bar{g}^2_{k,(i)} - \bar{g}^2_{k,(i)}.
\end{aligned}
\tag{23}
$$

These track local curvature proxy $a^{(i)}$, local minimizer proxy $b^{(i)}$, and gradient-noise variance proxy $\Sigma^{(i)}$.

Discrete-Time Element-wise Control Law

Using (23) with $m^{(i)} \approx \frac{1}{2}a^{(i)}(x^{(i)} - b^{(i)})^2$, the estimated discrete control is (for each $i = 1, \ldots, d$):

$$u^*_{k,(i)} = \begin{cases} 1, & a_{k,(i)} \leq 0, \\ \min\Big(1,\ \dfrac{a_{k,(i)}\big(\bar{x}_{k,(i)} - b_{k,(i)}\big)^2}{\eta\, \Sigma_{k,(i)}}\Big), & a_{k,(i)} > 0. \end{cases}$$

This produces an adaptive per-coordinate learning-rate scale.

Controlled SGD Update (Per Coordinate)

With the per-coordinate control $u_{k,(i)}$, the update is

$$x_{k+1,(i)} = x_{k,(i)} - \eta\, u_{k,(i)}\, \partial^{(i)} f_{\gamma_k}(x_k), \qquad i = 1, \ldots, d.$$

In practice, the control itself is also exponentially smoothed (same EMA philosophy) to avoid rapid oscillations.

This yields a robust adaptive SGD variant driven by:

- SME-based dynamics,
- optimal-control feedback law,
- local quadratic/diagonal approximation updated online.

Momentum SGD: coupled discrete dynamics

We consider *SGD with momentum* (MSGD) as a coupled recursion:

$$v_{k+1} = \mu v_k - \eta f'_{\gamma_k}(x_k), \qquad x_{k+1} = x_k + v_{k+1}. \tag{24}$$

- $\eta > 0$ is the learning rate.
- $\mu \in [0, 1]$ is the momentum parameter.
- γ_k is the (random) index of the sampled component loss at step k.
- Intuition: v_k is a "memory" of past update directions, which can accelerate progress along directions where the raw gradient steps would become small or oscillatory.

We will study MSGD by constructing a *stochastic modified equation (SME)* whose Euler scheme matches the one-step statistics of (24) at the relevant order.

SME viewpoint for momentum: introduce continuous variables

We introduce continuous-time processes (X_t, V_t) intended to track (x_k, v_k) on the time scale

$$t \approx k\eta.$$

The SME will be a coupled SDE:

$$\boxed{\begin{aligned} dV_t &= (\text{drift depending on } V_t \text{ and } X_t)\, dt + (\text{noise})\, dW_t, \\ dX_t &= (\text{drift depending on } V_t)\, dt. \end{aligned}}$$

Guiding idea:

- V_t encodes the *fast* velocity-like variable.
- X_t is the *position/parameter* variable.
- The momentum parameter μ will appear as a *damping* term in the V_t-equation.

SME for MSGD: coupled SDE

The SME corresponding to (24) is:

$$\begin{aligned} dV_t &= \Big(-\eta^{-1}(1-\mu)V_t - f'(X_t)\Big)\, dt \big(\eta\Sigma(X_t)\big)^{1/2}\, dW_t, \\ dX_t &= \eta^{-1} V_t\, dt. \end{aligned} \tag{25}$$

- The term $-\eta^{-1}(1-\mu)V_t$ is the *momentum damping*:

$$\text{effective damping} \quad \propto \quad \frac{1-\mu}{\eta}.$$

- The forcing term $-f'(X_t)$ matches the mean direction of descent.
- The diffusion term $(\eta\Sigma(X_t))^{1/2}dW_t$ matches the stochastic gradient variability.

Heuristically, (17) is a "noisy heavy-ball" dynamics with η-dependent scaling.

Quadratic test problem to isolate the role of μ

To understand *precisely* what momentum does, specialize to the 1D quadratic:

$$f(x) = \frac{1}{2}a(x-b)^2, \qquad \Sigma(x) \equiv \Sigma > 0 \ \text{ constant.}$$

Then

$$f'(x) = a(x-b).$$

Goal:

- quantify the *average descent rate* as a function of μ,
- quantify the *steady-state fluctuations* as a function of μ,
- see explicitly the *tradeoff* between speed and noise amplification.

A closed moment system for the quadratic case

Define a 3-dimensional moment vector:

$$M_t = \Big(\mathbb{E}\, f(X_t),\ \mathbb{E}\, V_t^2,\ \mathbb{E}\big(V_t f'(X_t)\big)\Big) \in \mathbb{R}^3.$$

Applying Itô's formula to the coupled SDE (25) yields a linear ODE system:

$$\dot{M}_t = A(\mu) M_t + B.$$

Explicit $A(\mu)$ and B for the quadratic moment ODE

For the quadratic $f(x) = \frac{1}{2}a(x-b)^2$ with constant Σ, the system (18) has:

$$A(\mu) = \begin{pmatrix} 0 & 0 & a/\eta \\ 0 & -2(1-\mu)/\eta & -2 \\ -2 & 1/\eta & -(1-\mu)/\eta \end{pmatrix}, \qquad B = \begin{pmatrix} 0 \\ \eta\Sigma \\ 0 \end{pmatrix}. \tag{26}$$

Interpretation of the three components:

- $\mathbb{E}f(X_t)$: average objective value.
- $\mathbb{E}V_t^2$: velocity energy (also tracks stochastic oscillations).
- $\mathbb{E}(V_t f'(X_t))$: coupling term controlling whether motion aligns with descent.

Stability regimes: concave vs convex quadratic

The sign of a changes everything:

If $a < 0$ (concave):

- $A(\mu)$ has a positive eigenvalue, so M_t diverges exponentially.
- Since f is negative, $\mathbb{E}f(X_t)$ decreases exponentially for all μ.
- The descent rate is maximized at $\mu = 1$.

If $a > 0$ (convex):

- for $\mu < 1$, all eigenvalues of $A(\mu)$ have negative real parts,
- hence M_t decays exponentially with a rate dominated by the eigenvalue with the *least negative* real part.

Dominant decay rate via the critical eigenvalue

For $a > 0$ and $\mu < 1$, the exponential decay of M_t is controlled by

$$\Re(\lambda(\mu))|,$$

where R denotes real part and

$$\lambda(\mu) = -\frac{1}{\eta}\Big[(1-\mu) - \sqrt{(1-\mu)^2 - 4a\eta}\Big].$$

Key qualitative consequences:

- Larger $\Re(\lambda(\mu))| \Rightarrow$ faster decay of the moments $\Rightarrow$ faster average descent.
- When the discriminant $(1-\mu)^2 - 4a\eta$ becomes negative, $\lambda(\mu)$ becomes complex $\Rightarrow$ oscillations appear.

Optimal momentum threshold for fastest (non-oscillatory) decay

The decay rate $|\Re(\lambda(\mu))|$ is maximized at

$$\mu_{\text{opt}} = \max\left(1 - 2\sqrt{a\eta},\, 0\right) \tag{27}$$

- For $\mu \leq \mu_{\text{opt}}$ the dominant eigenvalue is real and

$$\lambda(\mu) \sim -\frac{2a}{1-\mu} \quad \text{(to leading order in } \eta\text{)}.$$

- For $\mu > \mu_{\text{opt}}$, $\lambda(\mu)$ becomes complex and oscillations set in.

So momentum *helps* early descent (increasing rate), but too much creates oscillations.

Steady state and the fluctuation amplification effect

When the steady state is stable (convex $a > 0$, $\mu < 1$), the moment ODE (18) yields

$$M_t \to M_\infty = -A(\mu)^{-1}B = \begin{pmatrix} \dfrac{\eta\Sigma}{4(1-\mu)} \\ \dfrac{\eta^2\Sigma}{2(1-\mu)} \\ 0 \end{pmatrix}.$$

Two important takeaways:

- The leading-order magnitude of fluctuations scales like

$$|M_\infty| = O\big((1-\mu)^{-1}\big).$$

- Hence, increasing momentum ($\mu \uparrow 1$) *amplifies* the eventual noise floor.

This is the central tradeoff: **faster initial decrease vs larger steady-state variance**.

A scalar "averaged" model for the magnitude of fluctuations

Rather than controlling the full vector ODE (26), we approximate the evolution (in an averaged sense) by a scalar ODE:

$$\dot{m}_t = \Re(\lambda(\mu))\big(m_t - m_\infty(\mu)\big), \tag{28}$$

where $m_t \in \mathbb{R}$ represents the magnitude of M_t, and

$$m_\infty(\mu) = \frac{\eta\Sigma}{4(1-\mu)}$$

is the leading-order estimate of the steady-state magnitude.

Interpretation:

- $\Re(\lambda(\mu))$ controls the *speed* of relaxation.
- $m_\infty(\mu)$ controls the *noise floor* (larger when μ is larger).

Optimal control problem for μ_t in the scalar model

We now treat the momentum parameter as a (time-varying) control:

$$\mu = \mu_t \in [0, 1].$$

Control objective:

$$\min_{\mu:[0,T]\to[0,1]} m_T \qquad \text{subject to} \qquad \dot{m}_t = R\lambda(\mu_t)\big(m_t - m_\infty(\mu_t)\big).$$

This formalizes the practical desire:

- choose large momentum when it accelerates descent,
- reduce momentum when fluctuations dominate and you approach the noise floor.

Resulting feedback policy for adaptive momentum

Solving the control problem for the scalar model yields an (approximate) feedback law:

$$\mu_t^* = \begin{cases} 1, & a \leq 0, \\ \min\Big(\mu_{\text{opt}},\ \max\big(0,\ 1 - \dfrac{\eta\Sigma}{4m_t}\big)\Big), & a > 0, \end{cases} \tag{29}$$

with μ_{opt} given by (27).

Interpretation (convex case $a > 0$):

- If m_t is large (far from optimum), then $1 - \eta\Sigma/(4m_t) \approx 1$, so the rule selects $\mu_t^* \approx \mu_{\text{opt}}$ (fastest descent).
- If m_t becomes comparable to the noise scale, the rule reduces μ_t^* to control fluctuations.

Summary: what the quadratic analysis teaches us about momentum

For the MSGD dynamics (24) and its SME (25), the quadratic example yields:

- **Acceleration:** for $\mu \leq \mu_{\text{opt}}$,
$$\lambda(\mu) \sim -\frac{2a}{1-\mu},$$
so any nonzero momentum improves the initial convergence rate (convex case $a > 0$).
- **Oscillation threshold:** choosing $\mu > \mu_{\text{opt}}$ induces complex eigenvalues $\Rightarrow$ oscillatory behavior.
- **Noise amplification:** the steady-state magnitude scales like
$$m_\infty(\mu) = \frac{\eta\Sigma}{4(1-\mu)},$$
so higher μ increases the eventual fluctuations.
- **Control principle:** keep μ near μ_{opt} early, then lower it as you enter the fluctuation-dominated regime.